



Deep eutectic solvent liquid phase microextraction, powered by ultrasonic system, for determination of β -carotene in food samples

Nebiye Kizil^{a,b,c}, Buse Bozan Beydagi^b, Mehmet Lutfi Yola^d, Mustafa Soylak^{e,f,*}

^a Hasan Kalyoncu University, Faculty of Engineering, Department of Basic Sciences, Gaziantep, Türkiye

^b Hasan Kalyoncu University, Environmental Research and Application Center, Gaziantep, Türkiye

^c Hasan Kalyoncu University, Environmental Science & Energy Management Program, Gaziantep, Türkiye

^d Hasan Kalyoncu University, Faculty of Health Sciences, Department of Nutrition and Dietetics, Gaziantep, Türkiye

^e Department of Chemistry, Science Faculty, Erciyes University, Kayseri 38039, Türkiye

^f Turkish Academy of Sciences (TUBA), Cankaya, Ankara, Türkiye

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ABSTRACT

The research work is mainly concerned with the extraction of β -carotene, which is a terpenoid compound, from food and fruit samples using deep eutectic solvent based liquid phase extraction. Deep eutectic solvent liquid phase microextraction method, powered by ultrasonic mixing, (DES-UP-LPME) was carried out with deep eutectic solvent (DES) consisting of tetrabutylammoniumbromide (TBAB) and decanoic acid (DA). Key parameters such as pH, sample volume, tetrahydrofuran (THF) volume, DES type and volume, matrix effect, ultrasonication/centrifugation time were optimized to increase the extraction efficiency of β -carotene. Quantitative recoveries were obtained when pH, TBAB:DA ratio and sample volume were 3, 1:3 and 30, respectively. The limit of detection (LOD) and limit of quantification (LOQ) were found to be 0.005 $\mu\text{g/mL}$ and 0.015 $\mu\text{g/mL}$, respectively. Various food samples were analyzed spectrophotometrically by using the developed DES-UP-LPME at 450 nm. The addition/recovery studies were also performed on water samples to evaluate the accuracy of the method.

1. Introduction

Bioactive compounds are compounds found in food, animals or plants that have an effect on the body's health when consumed. Plant sources are the foods richest in bioactive compounds. These compounds are commonly found in functional foods, beverages, personal care products and dietary supplements. (Fernandes et al., 2019). The main bioactive compounds are prebiotics, probiotics, minerals, amino acids, phenolic compounds, vitamins and carotenoids. Carotenoids commonly found in the literature are: δ -carotene, γ -carotene, phytofluene, phytoene, zeacarotene, lycopene, neurosporene, α - and β -zeacarotenes, α - and β -carotenes. β -carotene is the most potent member of the highly pigmented carotenoids and is a major component of vitamin A. It is fat-soluble, highly bioactive and has antioxidant properties (Wang et al., 2021). Because of these properties, it has a wide range of applications in the pharmaceutical industry. It is used in pharmacological products as a food supplement with its strong antioxidant side (Bogacz-Radomska and Harasym, 2018).

β -Carotene, composed of carbon and hydrogen, is a secondary metabolite synthesized by plants (Bogacz-Radomska and Harasym, 2018). It has the same structure as the retinol molecule (Clark, 2007). The most common sources of β -Carotene are carrots, pumpkin, asparagus, peppers and peaches. However, it is possible to produce β -Carotene, which is a valuable source of vitamins and pigments, chemically, physicochemically and biochemically (Bogacz-Radomska and Harasym, 2018). Carotenoids with brightening pigments are frequently used in the food industry with these properties. β -Carotene, which is a predominantly yellow-orange color pigment, is highly preferred as an additive. It is particularly favoured in beverages due to its dominant color property. This component, which is suitable as a coloring agent, is used in the pharmaceutical industry, especially in syrups and tablets (Bogacz-Radomska and Harasym, 2018). The recommended intake of β -carotene by EFSA is 15 mg/day (EFSA, 2012; Zhang et al., 2022).

In order to determination of bioactive compounds diverse analytical techniques such as UV-vis spectrophotometry, high-performance liquid chromatography (HPLC), gas chromatography (GC), capillary

* Corresponding author at: Department of Chemistry, Science Faculty, Erciyes University, Kayseri 38039, Türkiye.

E-mail address: msoylak@gmail.com (M. Soylak).

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electrophoresis (CE) and also, fluorescence spectrophotometry is utilized (De Souza Dias et al., 2013; Michalkiewicz et al., 2008; Ozturk et al., 2018; Leite et al., 2021; Wu et al., 2012; Zu et al., 2006).

The analysis of β -carotene and traces species without enrichment is very difficult due to the interference effect and low concentration (Kanberoglu et al., 2019; Abbas et al., 2023; Isik et al., 2023; Isalou et al., 2023). Therefore, in addition to traditional methods for separation/preconcentration, techniques such as dispersive liquid-liquid microextraction, solid phase microextraction, liquid phase microextraction have been used within the scope of green chemistry in recent years. The application of environmentally friendly analytical processes and the use of chemicals has been a top priority focus of researchers since the 21st century due to the safe needs of community for ecology (Abbott et al., 2003; Abbott et al., 2017; Zhang et al., 2022). Therefore, the choice of solvent must be non-toxic in extraction of bioactive compounds. Green solvents such as supramolecular solvents, deep eutectic solvents, switchable natural deep eutectic solvents are used in dispersive liquid-liquid microextraction. In this study, one of the green solvents like a deep eutectic solvent (DES) was utilized for analyzing β -carotene.

DES, alternative solvents, has a low toxicity solvent system designed by combining two or more substances with different physical and chemical properties for the extraction of analyte (Ribeiro et al., 2015; Soyak and Koksall, 2019; Stupar et al., 2021). They are prepared via hydrogen bond acceptors (HBA) and hydrogen bond donors (HBD) by mixing at low temperatures. DESs are generally combined with quaternary ammonium salts (as HBA) and HBs such as carboxylic acids, amines and also, carbohydrates (Boateng, 2023; Martins et al., 2018; Sportiello et al., 2022). Besides their use in metalworking applications, DESs are also used in both liquid-liquid extraction, synthesis processing and CO₂ gas capture. Recently, DESs, consisting of hydrophobic non-toxic components and minimizing solvent consumption, have received more and more attention (Abbott et al., 2003, 2017; Smith et al., 2014). Recent studies have shown that many deep eutectic solvents (DESs) synthesized by choline chloride and organic acids are more efficient in extraction compared to conventional solvents. The most commonly used DES's are non-toxic quaternary HBA-based choline chloride. DESs have become the trend for green chemistry with advantages such as biodegradability, less toxicity and low cost (Alam et al., 2021).

In the present research, a simple, fast, inexpensive and effective ultrasonic-assisted deep eutectic solvent-based dispersive liquid-phase microextraction (UA-DLLME-DES) integrated with UV–Vis Spectrophotometry for the determination of β -carotene in some drugs, foods, tablets of vitamins samples.

2. Experimental

2.1. Instrumentation

The distilled water used in analysis and optimization studies was acquired from water distiller (Nüve ND 4 Water Distiller, Ankara, Turkey). In the DES-UP-LPME method, Thermomac brand vortex mixer was utilized to prepare the solutions quickly and easily. β -carotene was determined from samples using a Perkin-Elmer Lambda 25 UV–vis spectrophotometer (Perkin-Elmer, Norwalk, CT, USA). The pH measurements of the samples were performed with a pH meter (Hanna HI 2211, Woonsocket, RI, USA). J.P. Selecta (Spain) brand ultrasonic bath is used for ultrasound-powered liquid phase microextraction (JP Selecta™ Stainless Steel Ultrasonic Cleaning Baths w/o Heating, Barcelona, Spain). Its internal tank is made of AISI 304 stainless steel and has a switch for the operation of the heating from 0 to 80 °C. In addition, the ultrasonic bath is equipped with a timer for 0–99 min or continuous operation.

2.2. Reagents

The stock solution of β -carotene was prepared as 1.0×10^{-5} M via ultrasound and vortex mixer. This solution was diluted and used for optimization studies of the DES-UP-LPME method. Tetrabutylammoniumbromide (TBAB), choline chloride (ChCl) and decanoic acid (Da) were utilized for the DES combination and were obtained from Sigma-Aldrich (USA). The DES formation mechanisms of urea (Isolab Chemicals (Wertheim, Germany), and sucrose (Karlsruhe- Germany) were also investigated to examine the effect of different DESs on the extraction efficiency of the method. Tetrahydrofuran (THF), an aprotic solvent, was purchased from Merck, Darmstadt, Germany. Acetate, ammonium and phosphate buffers were prepared for pH adjustments. All chemicals have been used of analytical grade.

2.2.1. DESs preparation

In this study, DES was synthesized by the heating method. This was chosen because the literature review showed that it was the synthesis method with the highest yield, as well as being easy and fast to perform.

TBAB: Da, ChCl: urea and ChCl: sucrose, betaine: sucrose were mixed in different ratios of two different DES to prepare DESs consisting of HBA and HBD. The TBAB/Da DES solvent mixture with the highest recovery value, combined in 1:1, 1:2, 1:3 ratios were heated at low temperatures (30–90 °C). It was stirred by magnetic stirring under temperature control at 40 °C until a homogeneous and colorless solution was formed. (Chen et al., 2019; Kanberoglu et al., 2019). It was cooled at room temperature and used in β -carotene extraction (DES-UP-LPME method) studies.

2.3. UA-HDES-DLLME procedure

The model solutions were initially prepared in 50 mL tubes containing 5.37 μ g β -carotene and 1 mL buffer solution (phosphate buffer) for the application of the DES-UP-LPME method. The pH of the solutions was adjusted to 3 using 0.25 M NaOH and HCl solutions and then 100 μ L of quantitative type DES was added to them. 75 μ L of THF was used to separate the DES phase containing the analyte from the aqueous phase. The prepared model solutions were then kept in an ultrasonic bath for 7 min in order to allow the extraction process to take place the aid of ultrasound and to disperse the DES homogeneously into the aqueous phase. The ultrasonic wave broke the collected DES droplets into smaller droplets, resulting in a cloud-like solution and accelerating the extraction of the analyte.

The analyte which was extracted via DES and THF, was rapidly separated from the aqueous phase to the DES phase after centrifugation. After the addition of 1 mL ethanol to the final solution, β -carotene absorbance measurements were performed at 450 nm utilizing a UV–vis Spectrophotometer.

2.4. Sample preparation

Food, beverage, and water samples were analyzed by means of the DES-UP-LPME method for the determination of β -carotene. Food samples like pudding, yogurt, beverage, chocolate, vitamin including β -carotene (E160) were purchased from markets and pharmacy in Gaziantep, Turkey. Pudding, yogurt, vitamin samples were weighed in 2 g. and then 10 mL ethanol was added to them. The mixtures were shaken until the natural colorant has completely passed into the aqueous phase. Mixtures containing beta-carotene were centrifuged to obtain a homogeneous liquid phase.

On the other hand, the MWAE method has been carried out with two different volume ratios solvents for extracting β -carotene from orange, and carrot samples, weighing as 1 g. At this stage, the effect of both 1:3 (v/v) and 1:9 (v/v) ethanol-water mixture on the extraction efficiency of the β -carotene was investigated under microwave irradiation. The MWAE method conditions were determined as 80 °C, 5 min, and 400 W

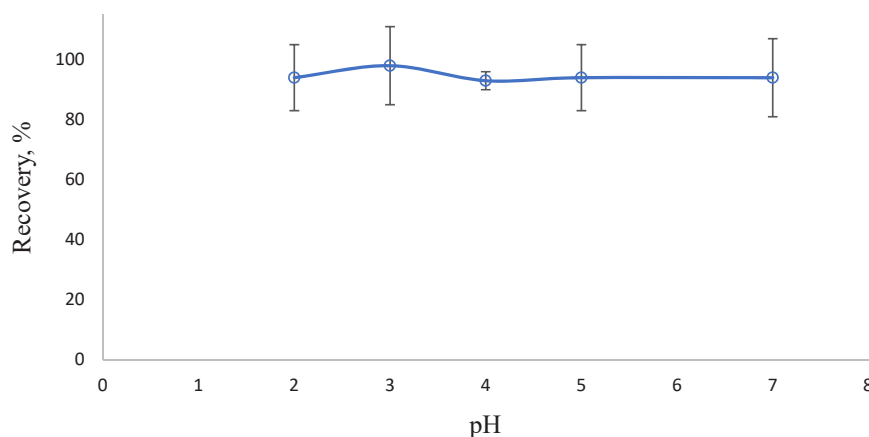


Fig. 1. Effect of pH on the extraction efficiency (pH:3, n: 3, eluent volume: 1 mL, deep eutectic solvent (DES) volume: 0.10 mL, tetrahydrofuran (THF) volume: 0.075 mL).

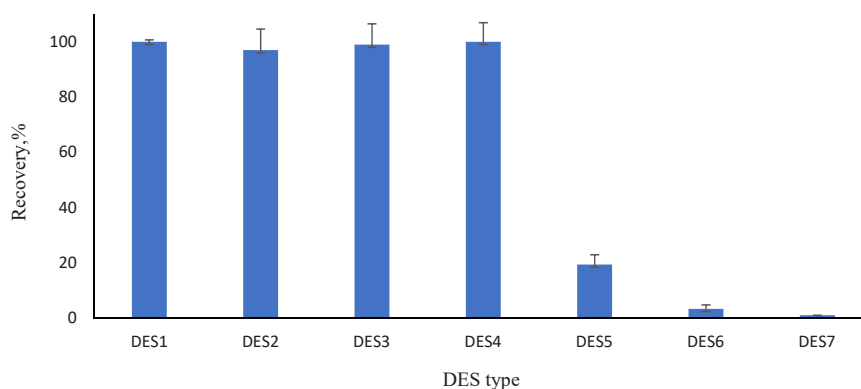


Fig. 2. Effect of deep eutectic solvent (DES) type on the extraction efficiency (pH:3, n: 3, eluent volume: 1 mL, deep eutectic solvent (DES) volume: 0.10 mL, tetrahydrofuran (THF) volume: 0.075 mL).

(Hiranvarachat et al., 2013). Besides, four types beverage samples, and milk were analyzed directly by using 0.1 mL. Eventually, the developed DES-UP-LPME method has been applied to the all-prepared samples.

3. Results and discussion

3.1. Effect of pH

The pH of the solution, which plays a critical role in the extraction process of the antioxidant food additive into the deep eutectic solution, should be optimized. Thus, the extraction efficiency of β -carotene is increased, thanks to the specific pH determined for each analyte. The influence of pH on the extraction of β -carotene from real samples was investigated in the range of 2–7 by adding buffer solutions. Recovery over 93 % was calculated at all values within the specified pH range, and the maximum extraction yield (98 %) was obtained at pH 3. It can result from electrostatic interaction between DES droplets and β -carotene molecules. At acidic and near neutral pH, electrostatic attraction occurs between positively charged β -carotene and negatively charged DES, facilitating the collection of the analyte. Recovery values of β -carotene were illustrated in Fig. 1 that the optimum pH was 3 for DES-UP-LPME method.

3.2. Influences of DESs type and volume

The main aim of these optimization experiments is to scrutinize the ability to extract more than 90 % of β -carotene from plants or some food samples. So as to obtaining a good extraction efficiency, we have devised

the DES-UP-LPME method. In this direction, DESs, is a green solvent, has been made up. Different DES combinations were prepared by using TBAB: Da (DES1), betaine: sucrose (DES5), ChCl: sucrose (DES6), ChCl: urea (DES7) and with a view to analyzing β -carotene. These compounds were selected to investigate different combinations such as quaternary ammonium salts and alcohol sugars. And, one of each type was synthesized to achieve the highest β -carotene extraction efficiency. It was explicitly seen from the Fig. 2 that the highest recovery values were gained when TBAB/Da DES solvent mixture was utilized.

The solvents, are HBA (TBAB), HBD (Da) and have been performed in the ratios (DES2)1:1, (DES3)1:2, (DES4) 1:3. In addition, for DES1 consisting of TBAB: DA with the best extraction efficiency, quantitative screening was carried out at ratios of 1:1, 1:2, 1:3 and these ratios were named DES2, DES3, DES4, respectively. Considering all the results shown in Fig. 2, it was decided that the feasible (DES4) ratio was 1:3. Therefore, the following parameters were carried out by adding DES4 to each of the prepared sample solutions.

The volume of DES determined was also studied in order to extract β -carotene completely from real samples and to analyze it correctly in the presence of matrix components. Hence, in the volume range of 100–300 μ L DES4 was added to the samples prepared according to aforementioned DES-UP-LPME method. The optimal DES4 volume was chosen as 100 μ L (Fig. S1).

3.3. Influence of THF volume

Due to the fact that the volume of tetrahydrofuran (THF) may affect the composition of the organic phase formed as a result of the extraction,

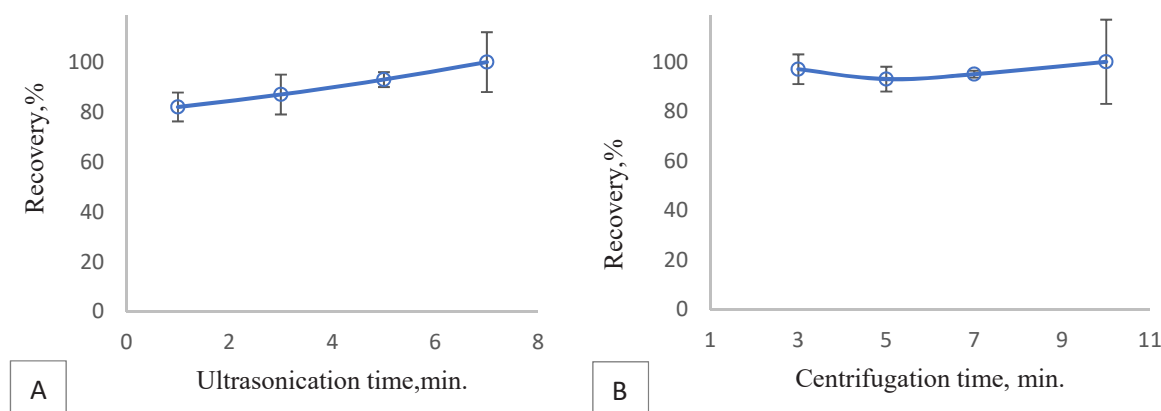


Fig. 3. The effect of sonication (A) and centrifugation time (B) (pH:3, n: 3, eluent volume: 1 mL, deep eutectic solvent (DES) volume: 0.10 mL, tetrahydrofuran (THF) volume: 0.075 mL).

the volume of the aprotic solvent was delved into in this parameter. In addition, self-aggregation occurs in the presence of an aprotic solvent and as a result of the interaction between the aprotic solvent and water (Aydin et al. 2017; Khezeli et al., 2015; Shishov et al. 2017). Taking all these into consideration, THF has been added to the all of sample solutions 50–150 μ L increasing volumes for extraction of β -carotene. According to results, quantitative recoveries were obtained in case of using 75 μ L. The recoveries started to decrease after 75 μ L of THF. This is because the composition of the organic phase formed as a result of the extraction probably deteriorated as the volume of THF increased. This result means that 75 μ L of THF is sufficient both to reduce the interactions between TBAB/Da: water molecules in the aqueous phase and for the analyte to pass into the DES phase.

3.4. Sonication and phase separation

Various techniques such as microwave irradiation, ultrasound-assisted have been used to obtain a turbid solution. In this paper, the ultrasound-assisted method shortened as DES-UP-LPME was performed for the preconcentration and analysis of β -carotene. To this end, each of the samples applied to the developed DES-UP-LPME method has been kept in an ultrasonic bath for 1–7 min. When the recovery values were examined in Fig. 3A, over 95 % recoveries were obtained in 7 min ultrasonic time.

Centrifugation was performed to separate the DES phase from the water phase of the cloudy solution formed after both the addition of THF and the ultrasonication process (Aydin et al. 2017; Khezeli et al., 2015; Shishov et al. 2017). The centrifugation time required for complete phase separation was studied in the range of 2–10 min at 4100 rpm, RCF 2.819 $\times$ g. High recovery values were obtained after 3 min. The results were illustrated in Fig. 3B.

3.5. Sample volume

Sample volume plays an extremely important role in analyzes using the liquid-phase microextraction method. The sample volume could be determined up to the maximum feasible point in order to get lowest error and highest pre-concentration factor. Therefore, this parameter

was investigated in sample solutions prepared between 5 and 50 mL, keeping the other optimal circumstances described above. According to the results given in the Fig. S2, it was seen that quantitative recoveries up to 30 mL sample volume are obtained. It shows that this method is applicable in real samples up to a sample volume of 30 mL. Considering all of these values, the preconcentration factor was calculated to be 30 by using a final volume of 1.0 mL.

3.6. Matrix effects

This DES-UP-LPME method was necessary for both the extraction and separation of the β -carotene from the matrix medium in real samples. Because, variations in matrix composition, such as ionic increment or the presence of other organic species, may affect the accuracy, precision and selectivity of the method. In addition, matrix components can bring about signal intensity to decrease or increase (Soylak and Dogan, 1996; Ghaedi et al., 2016; Stutz, et al., 2015; Tunay et al., 2022). Therefore, various ions and food colorants have been scrutinized for their detrimental effects on the extraction of β -carotene.

The matrix components such as chlorophyll, sunset yellow FCF, rhodamine B, indigo carmine, eosin B, chromotrope FB, anthocyanin, ponceau 4R, Zn^{2+} , Mg^{2+} , and Ca^{2+} were added to the sample solutions at different concentrations applying devised method. Then, absorbance measurements of final solutions were performed via UV–vis spectrophotometer. Finally, the tolerance concentrations which have no observed disruptive effect on β -carotene recoveries was determined. It was understood that the DES-UP-LPME method has a high selectivity in the light of the results, were shown in Table S1.

3.7. Real samples applications

The optimized DES-UP-LPME method was applied to the samples (e. g. carrot, orange, food products, and water sample) in which β -carotene was extracted by different methods, explained in the Section 2.4. The mixing process with alcohol has been applied to the pudding, yogurt, vitamin samples due to the fact that they can be easily and completely dissolved by mixing them with alcohol. The β -carotene concentrations of other samples such as carrots and oranges were determined using the microwave process in the 1: 9 ethanol-water solution owing to higher extraction efficiency. Because, the extraction efficiency in 1: 9 ethanol-water solution was 4.1 times higher for orange and 1.3 times higher for carrot than 1: 3 ethanol-water solution.

The mentioned samples were analyzed directly and β -carotene concentrations were given in the Table S2.

Also, addition/recovery experiments have been done to the water sample to prove the accuracy of the method. The main purpose of the method is based on the recovery of the added analyte with the highest

Table 1

Addition/recovery studies (pH:3, n: 3, eluent volume: 1 mL, deep eutectic solvent (DES) volume: 0.10 mL, tetrahydrofuran (THF) volume: 0.075 mL).

Samples	Added (μ g mL ⁻¹)	Found (μ g mL ⁻¹)	Recovery, %
Tap water	0.0	N.D. ^a	–
	0.054	0.055 $\pm$ 0.01	102
	0.108	0.097 $\pm$ 0.00	90

^a N.D.: Not detected, ^b Mean $\pm$ standard deviations

Table 2

Analysis of β -carotene in real samples (pH:3, n: 3, eluent volume: 1 mL, deep eutectic solvent (DES) volume: 0.10 mL, tetrahydrofuran (THF) volume: 0.075 mL).

Samples	Found ($\mu\text{g g}^{-1}$)
Pudding1	2.52 ± 0.16^b
Pudding2	0.81 ± 0.21
Yogurt	0.40 ± 0.18
Pastil	2.10 ± 0.20
Draje chocolate1	0.99 ± 0.64
Draje chocolate2	0.95 ± 0.50
Draje chocolate3	1.02 ± 0.40
Draje chocolate4	1.14 ± 0.43
Draje chocolate5	1.24 ± 0.83
Vitamin	15.4 ± 0.90
Orange	6.40 ± 0.75
Carrot	1.54 ± 0.22
Found ($\mu\text{g mL}^{-1}$)	
Beverage1	0.43 ± 0.06
Beverage2	0.21 ± 0.07
Beverage3	0.04 ± 0.01
Beverage4	0.08 ± 0.02
Milk	0.15 ± 0.03

^b Mean $\pm$ standard deviation

efficiency. Therefore, the analyte was added to drinking water in a concentration range from 0 to $0.108 \mu\text{g mL}^{-1}$ was added to drinking water and subjected to the DES-UP-LPME method under optimum conditions. Finally, extraction phases, completed 1 mL, were analyzed by means of Uv-vis spectrophotometer. When the results were commented in Table 1, it was observed that recovery values in the range of 90–102, high accuracy, and good sensitivity are achieved.

3.8. Analytical performance of DES-UP-LPME method

Standard deviation and regression statistical analysis were applied to determine the DES-UP-LPME method analytical performance. The analytical performance of the advanced technique has been evaluated under control of the DES-UP-LPME method under optimum conditions by determining the precision, linear calibration curve, limits of detection (LODs), and limits of quantification (LOQs). The relative standard deviations (RSDs) were calculated using the S_d/C formula to be less than 2.1 %. (S_b: standard deviation, C: concentration). The DES-UP-LPME method is considered to have good precision. The correlation coefficient (R^2) was estimated as 0.997 with regression statistical analysis by creating a linear calibration curve. Furthermore, the calibration curve

has been obtained as $A = 4.89 C + 0.0388$ (A: Absorbance and C: Analyte concentration) via β -carotene standard solutions. The pre-concentration factor (PF) of the current study was also found as 30 by means of elution (1 mL) and higher sample solution (30 mL). The detection and quantification limits were determined as $0.005 \mu\text{g/mL}$, $0.015 \mu\text{g/mL}$ respectively, depending on the 11 replicate blank solutions (signal to noise ratios of 3 and 10).

3.9. Applications

The common sources of β -carotene, obtained by extraction and fermentation, are vegetables and microalga *Dunaliella salina* (Kyriakopoulou et al., 2015; Ribeiro et al., 2011). There are varieties of extraction methods like Soxhlet extraction, maceration, solvent extraction, distillation, ultrasound-assisted extraction (UAE), microwave-aided extraction, supercritical liquid extraction (MWAE). In this research paper, MWAE which is a green extraction method, was performed to extract the β -carotene from the above-mentioned samples (Kyriakopoulou et al., 2015; Wang and Weller, 2006). Three samples of each yogurt, pastil, vitamin chocolate, beverages, milk, fruit and vegetable and water were analyzed. It was clear from the Table 2 that the method was carried out successfully in the pudding, yogurt, chocolate, vitamin, and various beverage samples to extract and analyze of β -carotene (E160). In addition, recovery values of over 90 % were obtained in the addition-recovery studies, demonstrating the accuracy of the method (Table 1).

4. Conclusions

In this study, DES-UP-LPME method was investigated to analyzing and enrichment of β -carotene. Firstly, pH, DESs type and volume, sonication and centrifugation time, sample volume, THF volume have been optimized. β -carotene was recovered over 95 % using 75 μL of THF and 100 μL of DES (TBAB/Da at a ratio of 1:3) at pH3. After the DES-UP-LPME method optimization was completed, it was successfully applied to some food, beverage and fruit samples. On the other hand, the 90–100 % recoveries were obtained in standard addition experiments to the tap water. Furthermore, β -carotene extraction was performed in orange and carrot samples via microwave heating. It significantly reduced time consumption compared to convective heating and other conventional extraction methods. According to results, the proposed method offers advantages such as high efficiency, minimizing solvent consumption, short processing time, low environmental impact and toxicity.

The DES-UP-LPME method was compared with other studies related

Table 3

Comparison between the developed method and other procedure from the reported literature.

Method	Limit of detection	Relative standard deviation, %	Preconcentration factor	Recovery, %	Samples	Ref.
Column switching technique (HPLC)	$0.50 \mu\text{g L}^{-1}$	1.43–2.59	–	96.66–102.40	food supplements	Brabcová et al. (2013)
Liquid-liquid microextraction (UV-vis)	$0.04 \mu\text{g L}^{-1}$	4.85–7.92	–	93.6–101.5	fruit juices	Sricharoen et al. (2016)
Supercritical fluid chromatography (SFC) (MS/MS)	0.1 ng mL^{-1}	10.41–12.28	–	102.31–104.13	egg yolk	Miao et al. (2023)
Liquid-Liquid Microextraction (HPLC)	$0.002 \mu\text{g mL}^{-1}$	4.1	–	90–103	fruit juices	Li et al. (2019)
Solvent extraction (UV-vis)	5 ng mL^{-1}	4.8–6.6	–	67.8–98.8	vegetables and nuggets	Biswas et al. (2011)
Cloud point extraction (UV-vis)	0.01 mg/L	1.5	10	99.6–111.3	fruit juice	Safdarian et al. (2021)
Dispersive liquid-liquid microextraction (Uv-vis)	$2 \mu\text{g L}^{-1}$	1.08	–	96.6–99.8	fruits and vegetables	Sereshti et al. (2014)
Liquid-phase microextraction (Uv-vis)	$0.038 \mu\text{g/mL}$	2.23–3.51	–	90.2–101.8	egg yolk	Wang et al. (2016)
Liquid-Liquid Microextraction (Uv-vis)	$0.005 \mu\text{g/mL}$	2.1	30	90–102	yogurt, pastil, vitamin, chocolate, beverages, milk, fruit and vegetable	This study

to β -carotene extraction in the literature. It was understood from the Table 3 that the method has a low detection limit value and high recoveries. The method also allows extraction in a wide pH range. This superiority shows that the study is more applicable than other studies. The ability to extract B-carotene from vegetables in a time- and energy-saving manner is another advantage of this study over other studies. An examination of the studies listed in Table 3 shows that the real sample variety was greater in this study than others. In other words, it has been applied directly to both commercial foods and fruit-vegetable. Consequently, in this work, not only the extraction of b-carotene, but also the analysis of b-carotene in real samples under the influence of interferences and low concentrations was successfully performed.

CRedit authorship contribution statement

Nebiye Kizil: Methodology, Visualization, Supervision, Writing – review & editing. **Buse Bozan Beydagi:** Methodology, Visualization. **Mehmet Lütfi Yola:** Methodology, Visualization, Writing. **Mustafa Soy lak:** Methodology, Visualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jfca.2023.105807](https://doi.org/10.1016/j.jfca.2023.105807).

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